

Lab Report: 7

Name: Saarthak Sabharwal

Roll number: 2023102055

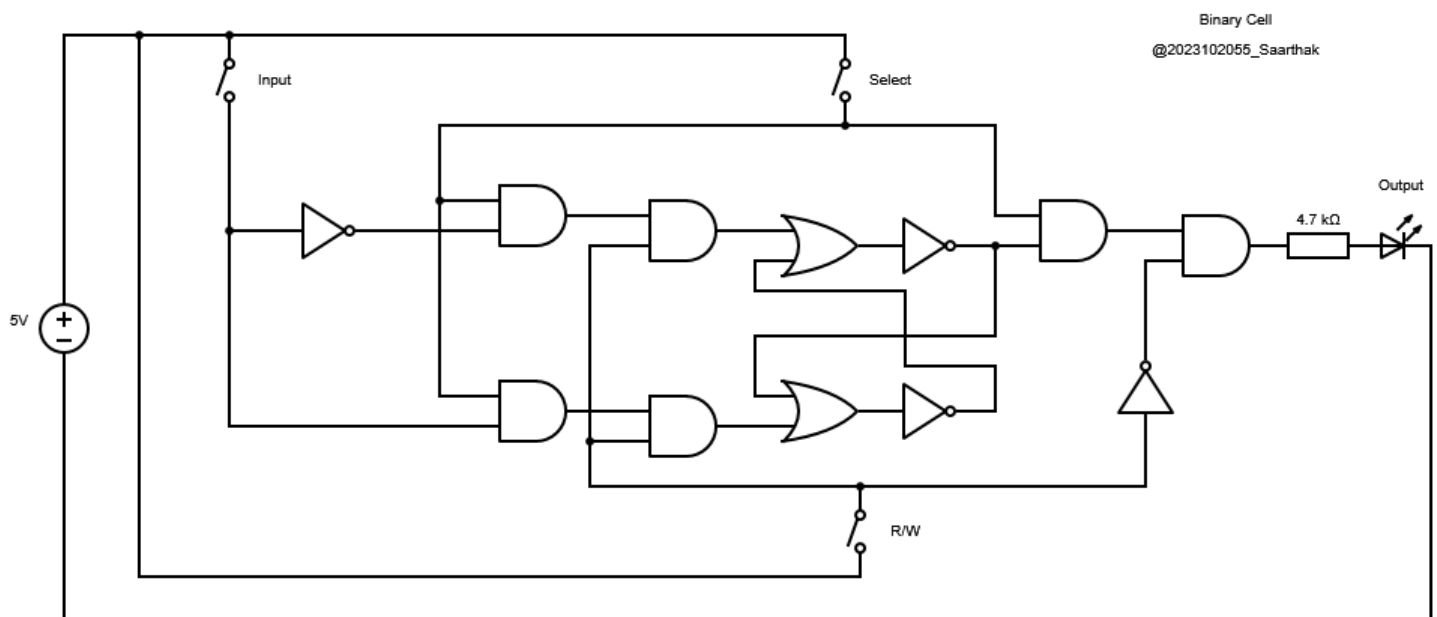
Group number: 6

Objective: To implement and verify the operation of a Binary cell for RAM based on RS flipflop.

Electronic components used:

- Digital test kit (w/ breadboard)
- 74HC04 (hex inverter) IC
- 74HC11 (triple 3-input AND) IC
- 74HC32 (quad OR gate) IC
- Wires

The reference circuit:



Procedure:

- Plug the ICs into the breadboard and power them up. Ensure that they all work correctly.
- Build the circuit as shown in the reference diagram.
- Test the circuit when the select switch is ON.
- Now, test the circuit with select switch OFF.

Conclusion:

The binary cell showed the following results as expected:

Select	R/W (0/1)	Input	Stored Bit (Q)	OUTPUT
0	0, 1	0, 1	Q	0
1	0	0	Q	Q
	0	1	Q	Q
	1	0	0	0
	1	1	1	0

Link for the Tinkercad simulation:

https://www.tinkercad.com/things/lBgZwDRIkLe-l7/editel?sharecode=ERguqgh5k0T3nJyfXpoY6_ethuCHanPGbFCmTutf7QA